

June 10, 2026

H. Van Dyke Parunak
Editor, *Autonomous Agents and Multi-Agent Systems*
Springer Nature

cc: Prof. Stephen Cranefield, Dr. Shuyue Hu,
Prof. Bastin Tony Roy Savarimuthu, Dr. Surangika Ranathunga
Guest Editors, Special Issue: “When Foundation Models Meet Multi-Agent Systems”

Dear Dr Parunak,

Please find attached the revised manuscript and a point-by-point response for Submission ID f265ac80-65ad-483a-bcfc-21d010e52041, prepared in response to your decision of 25 April 2026 inviting major revision.

Reframing. The revision recasts the contribution as an instantiation of the symmetric-stigmergic schema you formalize in *Insects and Agents: Extending the Metaphor* (MABS25). Foundation-model agents are deployed stigmergically, coordinating only through a shared artifact, and the paper maps each primitive of the schema concretely: the environment state es_i is a region’s pressure moving average, inhibition window, and provenance; the agent adjacency $aa(t)$ emerges from a two-stage validation gate; the environment-to-agent arrow ASU , absent from classic asymmetric stigmergy, is realized as band and model escalation when the field stalls; and the two pheromone stores realize ESD as evaporation. The method uses the three-operator vocabulary (aggregate, propagate, evaporate) of *Agents, Pheromones, and Mean-Field Models* (AAMAS11), and the related work is rebuilt as a single lineage from digital pheromones to language-model agents.

Your six points. Each is addressed substantively. The novel terminology is dropped for pheromone and stigmergy vocabulary, and the title changes accordingly. Pheromone propagation over the domain’s temporal structure (the ESP operator, one hop per tick over time-block adjacency) is now implemented in the kernel and ablated, answering the impoverished-structure concern directly. The manuscript is cut from 80 to 52 pages. The small problem scale is acknowledged plainly as a stated limitation rather than worked around. And on accurate reporting: a code audit found a system-prompt confound in the baselines and a pressure-metric mismatch across strategies, both now fixed, so every result derives from a complete regenerated 1350-trial grid on one authoritative metric, with each claim scoped or accompanied by a confidence interval.

The work fits the special issue “When Foundation Models Meet Multi-Agent Systems” through a bidirectional mechanism: foundation models supply the open-ended actions classic stigmergy could not enumerate, and symmetric-stigmergic coordination supplies the pressure-based criterion that combines those proposals and escalates model capability. The point-by-point response treats each item with pointers into the revised manuscript.

Sincerely,

Rolando Rene Rodriguez, Jr.